Google Cloud H2S

AI Builder Cup

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Build for the world.



Team Details

a. Team name: **[Your team name]**
b. Team leader name: **[Your full name]**
c. Problem Statement: **Retail & Commerce: Intelligent Customer and Business Experiences. Etalase makes WhatsApp & Instagram sellers discoverable and buyable by AI shopping agents**

Brief about the idea

Etalase turns a WhatsApp seller's product photos and chats into a storefront that AI shopping agents and ordinary shoppers can find and buy from, with every order landing back in the seller's chat.

Why now

AI shopping is arriving. Google's UCP checkout is live in the US, started with select retailers in Australia and is in limited test in India. Indonesia has no date; there, AI discovery is live but agent checkout is not. Morgan Stanley's US scenario: US$190–385B by 2030.

The problem

Agents only read structured catalogs. Across Southeast Asia, small sellers run on chat apps such as WhatsApp, LINE, Zalo and Messenger, with catalogs in photos and messages, so agents cannot see them. When agent checkout arrives, that demand flows to marketplaces and big brands instead.

Our solution

Gemini builds the catalog from photos, voice notes and spreadsheets. Etalase publishes it over MCP, the UCP checkout API and a store link that works today without an agent, reserves stock, takes QRIS payments and holds the payout through a return window. The seller confirms and ships from WhatsApp or Telegram.

US$185B

Southeast Asia e-commerce GMV in 2025; consumer interest in AI is 3× the global average

95%

of Indonesian e-commerce businesses use chat apps such as WhatsApp to sell; only 18% use marketplaces (BPS 2023, multiple answers)

1.8%

of Indonesian e-commerce businesses sell through their own website, so few have a structured catalog (BPS, 2023)

11%

of small and mid-size merchants are ready for AI shopping agents, vs 19% of large merchants (2026)

Sources: e-Conomy SEA 2025 (Google, Temasek, Bain, Nov 2025); BPS Statistik E-Commerce 2023 (pub. Jan 2025; businesses can use several channels); PYMNTS × Visa, 2026 Global Digital Shopping Index (Sep 2026, global vendor survey); Morgan Stanley, “Here Come the Shopping Bots” (US only). Links in the repo: docs/15-evidence.md.

Opportunities

How is it different?

- Agent-readiness tools (Era, ShopAgentic, AgentReady) serve Shopify-style stores that already have product data
- Marketplaces and platforms make sellers agent-ready only inside their own ecosystem, at their fees
- Chat tools (Meta AI, cshub, Mayar) serve human buyers; we found none that builds the catalog from photos
- **Etalase serves sellers with no store system: photos to catalog, a store link, MCP and UCP**

How does it solve the problem?

- Builds the catalog from what sellers already have: photos, voice notes, chats, spreadsheets
- Publishes it in agent-readable form (MCP tools, UCP checkout API, schema.org) and as a store link anyone can open
- Runs a safe checkout: stock reservation, seller confirmation, QRIS, courier rates, and a payout held until the return window ends
- Keeps the seller in WhatsApp (or Telegram), with no new app to learn

USP of the solution

- **From WhatsApp photos to an agent-ready catalog with Gemini, no store system needed**
- Brings chat-first micro-sellers into AI commerce with a channel they own, paid only per order
- One core with pluggable country packs for payment, shipping and OMS
- Privacy by design: tenant-isolated AI and minimal data, built with Indonesia's PDP Law in mind

List of features offered by the solution

AI catalog builder

Built

Gemini reads product photos, voice notes, captions and CSV files, then drafts products, variants, prices and weights for the seller to confirm.

Agent-ready storefront

Built

Each shop speaks MCP, UCP and schema.org, scores 0–100 for agent readiness, and has a mobile store link for buyers without an agent.

WhatsApp seller agent

Built

An ADK agent updates stock and prices and handles orders and tracking, on WhatsApp or Telegram. Buttons never go through the model.

Safe agent checkout

Built · QRIS mocked

Stock is reserved at checkout and low-confidence stock is confirmed with the seller first: 0 units oversold when 100 buyers race.

Shipping & payouts

Mocked in demo

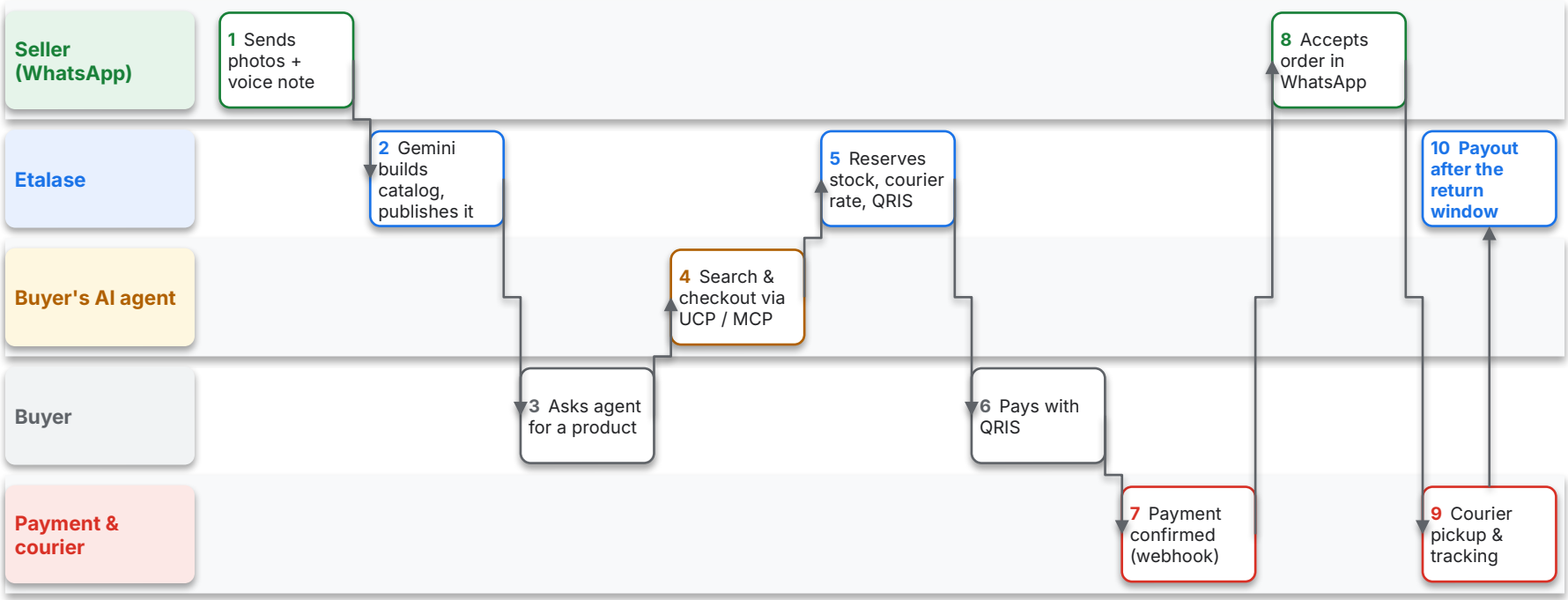
Courier rates; the payout waits out a return window, then is released minus commission. Biteship and Xendit are wired; the demo uses mocks.

Connectors

Built · more next

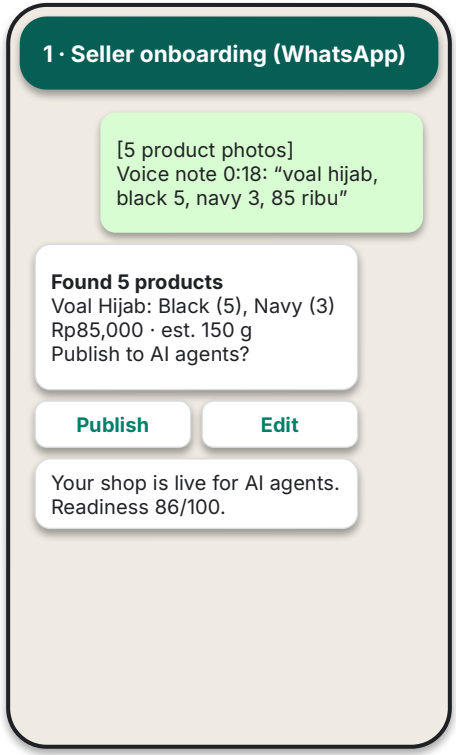
Instagram post import and an OpenAPI Connector API for website builders work today. Google Sheets, OMSs and WooCommerce come next.

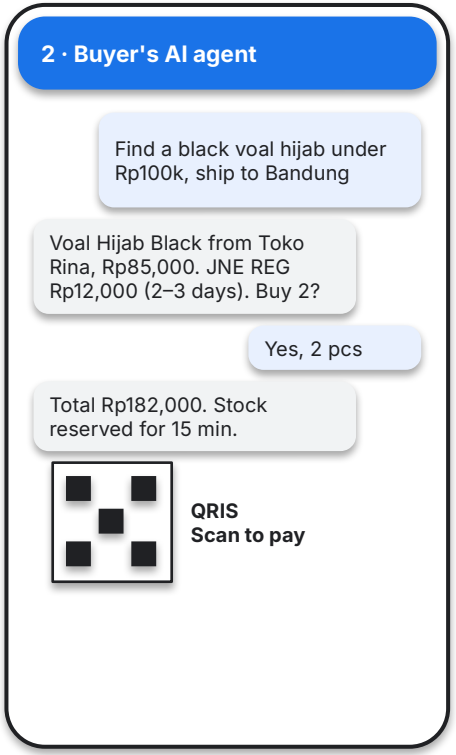
Process flow diagram or Use-case diagram

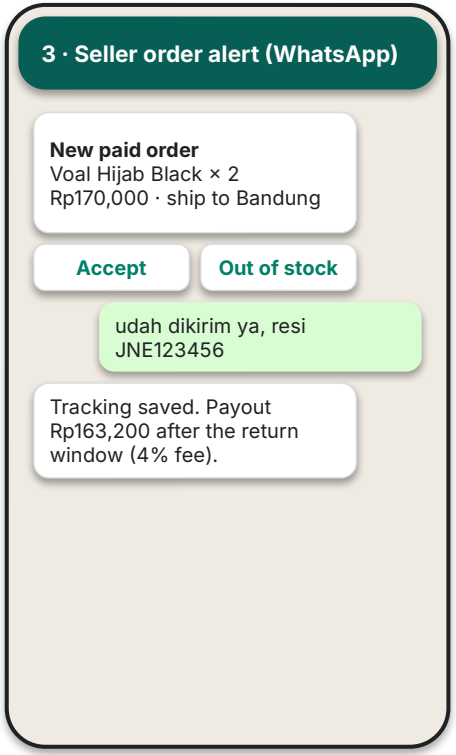


A buyer without an agent opens the store link and follows the same steps from 5. If stock confidence is low (stale or nearly sold out), the seller confirms stock before the QRIS link is sent. Stock reservations expire after 15 minutes if unpaid.

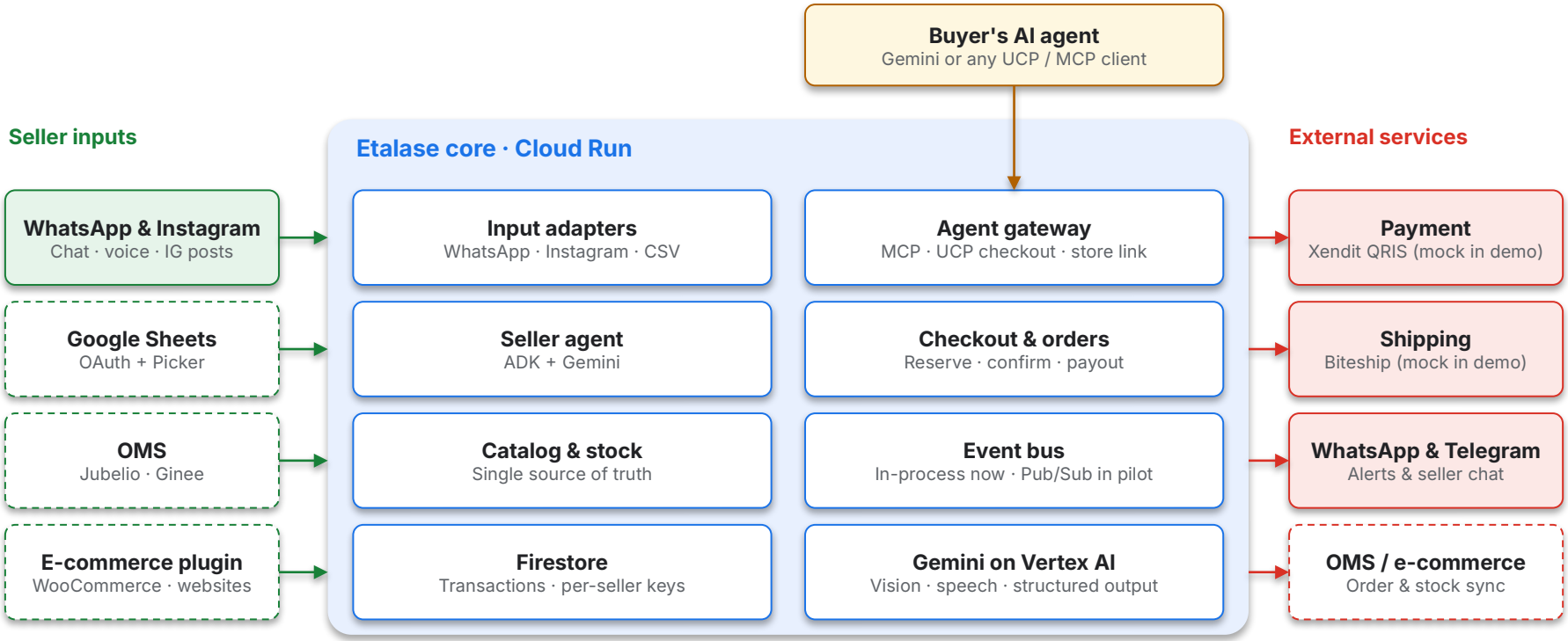
Wireframes/Mock diagrams of the proposed solution (optional)







Architecture diagram of the proposed solution



Solid = hackathon MVP · dashed = roadmap. Country pack = { currency, payment, shipping, tax } adapters: a new market adds one config entry, not a new core. Each seller has one stock owner: Etalase, the OMS or the store.

Technologies to be used in the solution

AI & agents

- Gemini on Vertex AI: photos, voice and text to catalog with structured output
- ADK seller agent with tenant-scoped tools
- ADK buyer agent calling our MCP tools (demo)

Agentic commerce

- MCP server: search, shipping, checkout, status
- UCP profile and checkout API (/ucp/v1)
- schema.org data and store link
- Next: Merchant Center feed, AP2 payments

Google Cloud runtime

- Cloud Run: API, MCP, webhooks, dashboard
- Firestore store with transactions
- Secret Manager for every key
- Pilot: Pub/Sub, Cloud Tasks, Storage, KMS

Integrations

- WhatsApp Cloud API and Telegram Bot API
- Xendit QRIS and Biteship adapters
- Instagram API (catalog import)
- OpenAPI Connector API

Seller dashboard

- Lightweight web dashboard on Cloud Run
- One-time-code sign-in over chat
- Readiness, orders, returns, payouts

Security & operations

- Tenant isolation, including AI context
- Signed webhooks, fail-closed startup checks
- Rate limits, OTP lockout, CSP
- Built with Indonesia's PDP Law in mind

Estimated implementation cost (optional)

Pilot assumption: 100 sellers · 3,000 products · 1,000 orders per month. Rough estimates, to be validated with the Google Cloud Pricing Calculator.

Component	Basis	Est. / month
Cloud Run, Firestore, Storage, Pub/Sub	Low pilot traffic, mostly within free tiers	US$10–30
Gemini on Vertex AI	~3,000 catalog extractions + ~2,000 seller-agent turns (buyers' agents run on their own models)	US$15–60
WhatsApp Cloud API	~4 templates + 3–5 charged replies per order × 1,000 orders at ~IDR 357–460	~US$160–260
Biteship	Grow plan from 2 Nov 2026: Rp149,000 per month	~US$9
Payment-platform account	Xendit xenPlatform: Rp25,000 per active seller per month (unconfirmed)	~US$155
QRIS payments	MDR 0% to Rp100k (all) or Rp500k (micro), else 0.7%; fixed gateway fee, if any, paid by buyer	Pass-through
Total platform cost		**≈ US$350–520**

Unit economics per order

≈ IDR 5–7.6k

platform cost per order at 10 orders per seller, accounts included

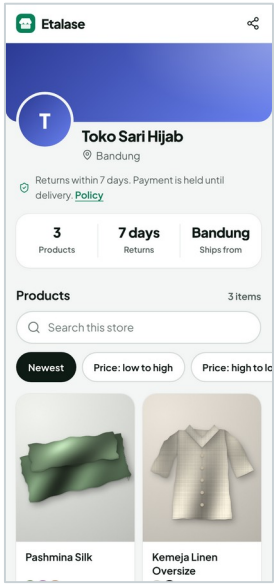
IDR 6.8k

commission at 4% on an IDR 170k order. Break-even is about 20 orders per seller per month, so we are testing a minimum fee.

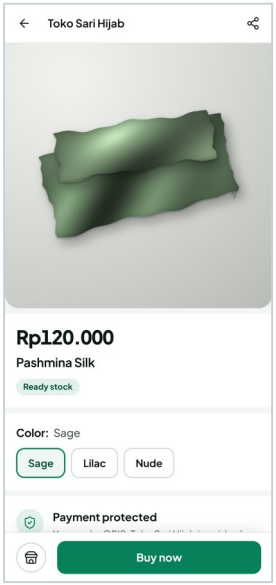
Price references: Meta WhatsApp pricing (service and in-window utility replies charged from 1 Oct 2026 per Meta's newer page, to be re-confirmed; Indonesian rate via Qontak); Biteship and Xendit price pages (account fee unconfirmed); Bank Indonesia QRIS MDR release; Vertex AI Gemini list prices. FX: US$1 ≈ IDR 16,000. Our arithmetic; Telegram has no per-message fee. Details: docs/10.

Snapshots of the prototype

Screenshots of the running prototype with mock payment and courier. Product photos are synthetic samples.



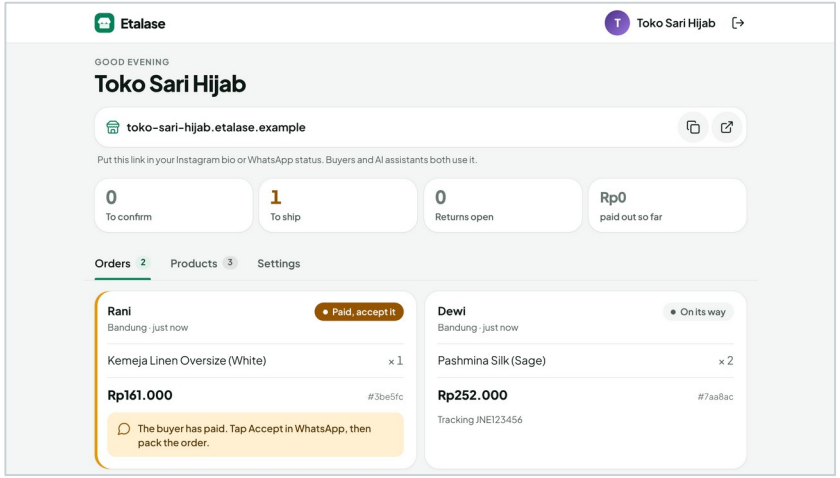
Store link



Product page



Pay with QRIS



Seller dashboard

Prototype Performance report/Benchmarking

439 automated tests pass today (stock races, payments, returns, webhooks, OTP abuse, fixes from an independent review) and an end-to-end run over real HTTP. TBD results come from the Gemini run on Cloud Run.

Metric	How we measure	Target	Result
Catalog extraction accuracy	50 labelled products: title, variant, price, stock	≥ 90% of fields correct	TBD
Catalog build time	Photos + voice note to published product	< 15 s per product	TBD
Agent tool latency	p95 of search_products and create_checkout	< 2 s	TBD
Overselling under load	100 concurrent reservations for the last unit	0 units oversold	Pass (in-memory)
Seller alert speed	Payment webhook to WhatsApp message	< 5 s	TBD
Duplicate & late payments	Same event twice; payment after expiry	No double change; auto refund	Pass
Returns and the payout hold	Refund before the payout; silent seller; courier-returned parcel	Refund never depends on the seller	Pass
Stock-update understanding	Intent accuracy on 100 real seller messages	≥ 95%	TBD

Additional Details/Future Development (if any)

1 2 3 4

Hackathon MVP

- WhatsApp, Telegram, Instagram, CSV
- MCP, UCP checkout, store links
- Demo buyer agent on Gemini (ADK)
- Payout hold and returns; mock QRIS and courier

Indonesia pilot

- 50–100 sellers onboarded
- Firestore, Pub/Sub, Cloud Tasks
- KYC and payouts via sub-accounts
- Sheets sync, Merchant Center feed

More channels

- OMS adapters: Jubelio, Ginee
- E-commerce plugins, WooCommerce first
- Shopee and TikTok Shop API sync
- Sales analytics for sellers

Regional

- Country packs: MY, PH, TH, VN
- More chat adapters: Instagram DMs, LINE, Zalo
- Stripe in supported markets
- Cross-border shipping and duties

What we will validate next

Ten seller interviews: willingness to pay and to wait for the payout, orders per month (break-even ≈ 20), return rates · whether the payment provider can enforce the payout hold, and its legal classification · whether buyers arrive through agents or the store link · UCP timing in Indonesia · extraction quality with Gemini

Provide links to your:

GitHub public repository	[Add GitHub link: github.com/<team>/etalase]
Demo video (3 minutes)	[Add YouTube / Drive link]
Final product link	[Add Cloud Run URL]

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Thank you

